2.2 Adding and Subtracting Rational Expressions – Lesson

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a) $\frac{1}{6}+\frac{1}{5}$

> ***Note:*** *the product of the denominators will give a common denominator (but not always the lowest common denominator)*

b) Simplify and state restrictions

$$\frac{x^2-1}{x^2-4}\times\frac{x^2+3x-4}{x^2+5x+4}$$

c) $\frac{2x}{3y}+\frac{y}{6x}$

Add and Subtract Rational Expressions With Monomial Denominators

a) $\frac{1}{5x}+\frac{1}{2x}$

b) $\frac{ab^2+2}{2ab^2}-\frac{b+2}{2b}$

Add and Subtract Rational Expressions with Polynomial Denominators

a) $\frac{x+5}{x-3}+\frac{x-7}{x+2}$

1. get a common denominator
2. expand numerators
3. add/subtract fractions
4. simplify where possible
5. state restrictions (throughout process)

b) $\frac{x+9}{x^2+2x-48}-\frac{x-9}{x^2-x-30}$

c) $\frac{x-2}{x+2}+\frac{x+10}{x^2+6x+8}$

d) $\frac{2x}{x-1}-\frac{x+2}{x^2+3x-4}$

e) $\frac{a+1}{5-2a}-\frac{a-4}{2a-5}$